

Photometric Stellar Classification Using Machine Learning data from SDSS

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Abstract

The classification of stars along the Harvard spectral sequence traditionally relies on spectroscopic observation, which is very costly. This study demonstrates that broadband photometry from the Sloan Digital Sky Survey, combined with supervised ML can recover spectral types with high accuracy. A sample of 8996 OBAFGKM stars were taken from the SDSS SpecPhoto cataloged using 5 dereddened magnitudes (u,g,r,i,z) and a $(g-r)$ color index as classification features. Class imbalance was addressed through stratified resampling, and a Random Forest classifier was trained on the resulting balanced dataset. A test accuracy of 88.43% and a cross validated accuracy of $89.2 \pm 1.2\%$ were achieved. Performance is perfect for the extreme O and B type stars, while the F–G–K classes present an intrinsic degeneracy that limits discrimination. Feature importance analysis confirms that the u and g bands are dominant discriminators, consistent with temperature sensitivity of the blue optical photometry. The results highlight the viability of photometric classification at a survey scale.

1 Introduction

The classification of stars by spectral type stands as one of the most enduring achievement of astrophysics. The Harvard spectral sequence- ordered O,B,A,F,G,K,M with decreasing

effective temperature- provides a systematic route to understanding atmospheres, ranging from hot, ionized plasmas of O type stars to the cool molecule rich photo-spheres of M type dwarfs [5]. Classically, this sequence is determined through spectroscopy, identifying characteristic absorption features such as the Balmer series in A-type stars or molecular bands in the K and M type stars [2]. However, obtaining resolved spectra of every observed star is expensive, and as modern sky surveys catalog hundreds of millions of objects, purely spectroscopic classification becomes increasingly impractical [6].

This motivates the question at the heart of the study: how accurately can spectral types be recovered from broadband photometry? Using data drawn from the SDSS one of the most comprehensive catalogs available- this report applies a Random Forest classifier to five band dereddened photometry to attempt automated spectral classification across the full Morgan-Keenan sequence. Beyond simply reporting accuracy, the aim is to understand where and why the model succeeds or fails to classify stars into their types.

2 Data

The study features photometric data captured by the Sloan Digital Sky Survey (SDSS) [1], which is one of the most comprehensive imaging and spectroscopic surveys to date. The dataset used was queried directly from the SDSS, SpecPhoto table using the astroquery Python library [3], which retrieved 10,000 stellar objects which were classified as STAR by the SDSS.

Every star in the dataset includes 5 distinct dereddened photometric magnitudes across the filter systems in the SDSS: u , g , r , i and z , spanning wavelengths from ultra-violet to near-infrared. Dereddened magnitudes correct for interstellar dust extinction, which helped ensure that the observed colors reflect intrinsic stellar properties rather than line of sight dust contamination.

After filtering for stars with valid OBAFGKM stellar classifications which removed entries with missing values, 8,996 stars remained after passing through the filter. The dataset still exhibited significant class imbalance, with F-type stars comprising 29% of

the entire sample and O and B-type stars took less than 1% combined. The imbalance reflects the stellar population distribution but would create a bias in a classifier towards common classes, hence this was addressed in the preprocessing as described in Section 3.

3 Methodology

3.1 Data Preprocessing

Prior to the model training, the original query was filtered through a series of preprocessing steps. Records were first filtered to only keep stars belonging to the seven primary Harvard spectral classification system which was developed by Annie Jump Cannon- O, B, A, F, G, K, M- using an expression match on the subClass field. The spectral class label was then reduced to its leading character which represented their primary spectral type. Any remaining rows which contained missing values were discarded.

3.2 Class balancing

A class imbalance has been exhibited in section 2. Hence, each spectral class was re-sampled to 500 samples of each class type using the resample utility from scikit-learn [4]. Classes with more than 500 samples were under-sampled without replacement, while classes with fewer than 500 were over-sampled with replacement. The outcome provided us with a balanced dataset of 3500 record with equal class representation across all seven types.

3.3 Feature Set

The final set consisted of six variables: the five dereddened SDSS photometric magnitudes (dered_u, dered_g, dered_r, dered_i, dered_z) and their derived color index, computed as the difference between the g and r band magnitudes ($g - r$), which serves as a proxy for stellar temperature. Together, these encode the spectral energy distribution from ultraviolet to near-infrared range.

3.4 Classification Model

A Random Forest Classifier was selected for this multi-classification task, which is an ensemble method [7] that pools predictions from a large number of independently trained decision trees, which leads to reduced variance and improves generalization. This is suited to the context for three reasons: it is very robust, it handles multi-class targets without requiring class binarization and it produces important scores as a by-product of training. The model was configured with 100 estimators and a fixed random seed (`random_state=42`) to ensure reproducibility.

The balanced dataset was partitioned into a training set (80%) and a held out test set (20%) using `train_test_split` with stratification, which ensured that the class distribution was preserved in both sets. This model was trained exclusively on the training partition.

3.5 Model Evaluation

Model performance was assessed using two methods. First, the classifier was evaluated on the held out test set, giving its overall accuracy along with its per-class precision, recall and F1-score. Second, 5-fold stratified cross-validation was performed on the entire set using the `StratifiedKFold`, which partitions the data into five folds and trains and judges the model on each fold. This ensures that the proportions are preserved. The mean and standard deviation of accuracy across the five folds provide a more reliable estimate on model performance. A confusion matrix was additionally produced to examine per-class misclassification patterns.

3.6 Visualization

Two visualizations were plotted to contextualize the results within an astrophysical framework. A Hertzsprung Russell diagram was plotted using the color index which was calculated beforehand ($g - r$) as a proxy on the horizontal axis and apparent magnitude (inverted, with brighter stars at the top) on the vertical axis. Each data point was rendered in an estimate blackbody color corresponding to its spectral class, placing the

classified stars within the classical stellar temperature-luminosity relationship. Additionally, a feature importance graph was generated from the trained Random Forest model to identify which bands contributed to the most decisions, providing interpretability alongside quantitative performance.

4 Results

4.1 Classification Performance

The Random Forest classifier achieved an overall accuracy of 88.43% on the held-out test set ($n=700$), which demonstrates strong discriminatory power across the entire Harvard spectral sequence. This result is broadly consistent with the 5-fold cross validation estimate of $89.2 \pm 1.2\%$, which confirms that the model generalises and is not sensitive to any particular train-test partition. The low standard deviation across folds indicates stable, reproducible performance.

Per-class performance metrics are reported in Table 1. The classifier achieves near-perfect discrimination for the hot, early-type stars: spectral classes O and B attain F1 scores of 1.00 and 1.00 respectively, which reflect their extreme photometric features which are characterized by very blue color indices ($(g-r) < 0$) which render them photometrically distinct from all other classes. Class A similarly performs strongly (F1=0.93), occupying a well-separated region of color space.

Performance degrades progressively toward the intermediate spectral types. Class K and G receive F1 scores of 0.88 and 0.80 respectively, while class F is the weakest performer at F1=0.67. This systematic trend is physically interpretable: F-, G-, and K-type stars occupy adjacent and overlapping regions which cause a blind spot of the stellar color-temperature sequence, with the $(g-r)$ index varying gradually across this range. The boundaries between these classes are therefore inherently ambiguous from photometric data alone, without recourse to spectral line diagnostics.

Table 1: *Per-class classification metrics on the held-out test set.*

Class	Precision	Recall	F1-Score	Support	Notes
O	1.00	1.00	1.00	100	Perfect separation
B	0.99	1.00	1.00	100	Near-perfect
A	0.94	0.92	0.93	100	Strong
M	0.93	0.90	0.91	100	Strong
K	0.86	0.90	0.88	100	Good
G	0.81	0.80	0.80	100	Moderate
F	0.67	0.67	0.67	100	Weakest
Macro avg	0.88	0.88	0.88	700	

4.2 Confusion Matrix Analysis

Inspection of the confusion matrix confirms the quantitative metrics. The dominant source of error is the mutual misclassification of F- and G- types of stars: a non-trivial fraction of F-type objects are assigned to class G, and vice versa. Errors only took place between adjacent classes. This is consistent with the overlap in photometric color between those two spectral types. They differ primarily in temperature by only ~ 1000 K- a distinction that the broadband SDSS filters are not sensitive to. Another confusion arises between G and K for analogous reasons. In contrast, the hot classes and the cool type stars exhibit negligible off-diagonal counts continuous with their photometric distinctiveness.

4.3 Feature Importance

The feature importance scores extracted from the trained model reveal that `dered_u` and `dered_g` bands contribute most strongly to classification, followed by `dered_r`. The color index $(g - r)$ also carries significant discriminatory power as a single composite feature. The redder bands contribute less, though they retain relevance for distinguishing cool M-type stars, whose spectral energy distributions peak at longer wavelengths. This ordering is physically consistent with the expectation that stellar effective temperature is most sensitively encoded in the blue to green portion of the optical spectrum.

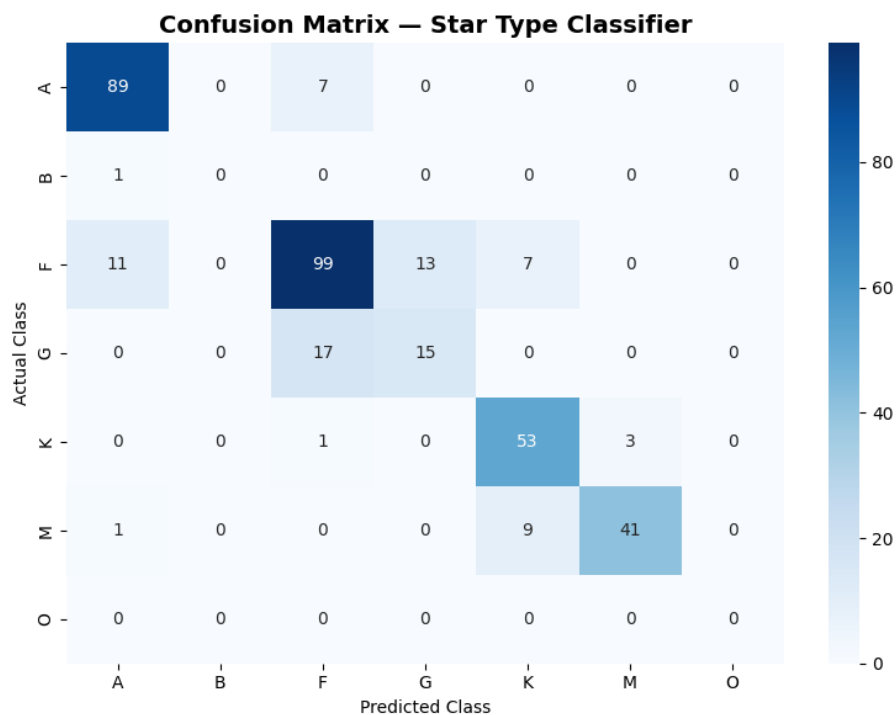


Figure 1: Confusion Matrix (Figure 3) for the held-out test set

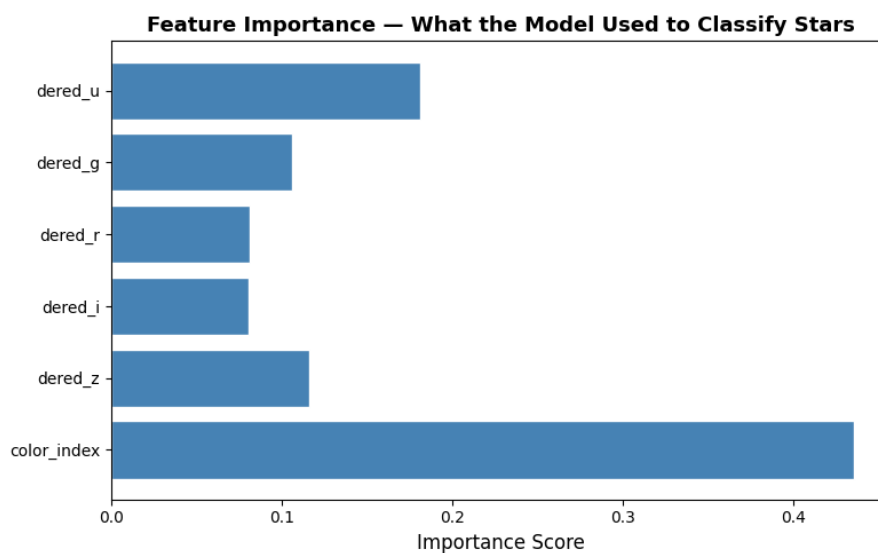


Figure 2: Feature Importance Graph

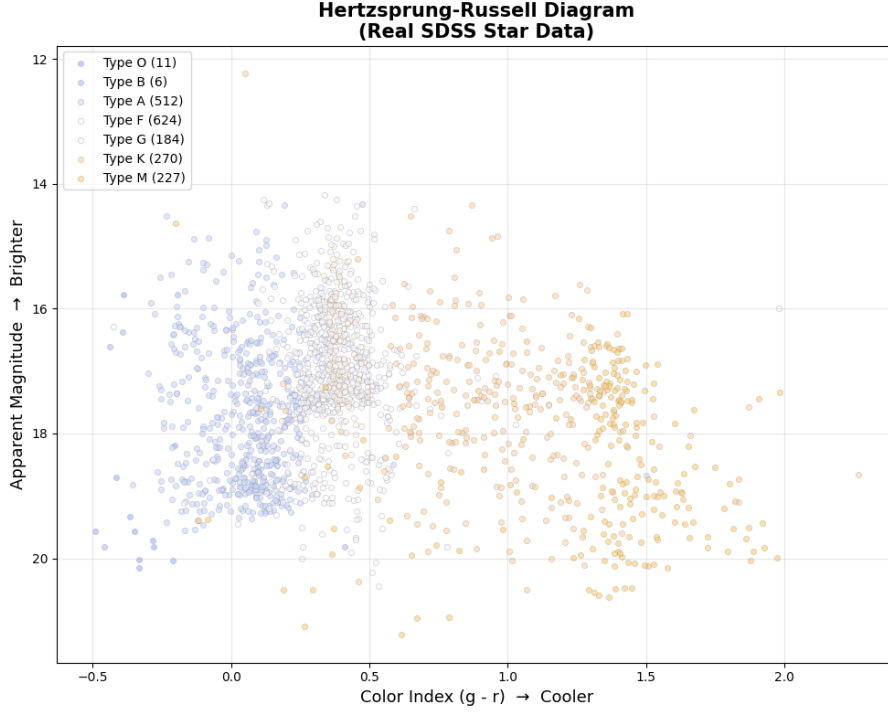


Figure 3: Color-Magnitude diagram

4.4 Hertzsprung-Russell Diagram

The HR diagram which was constructed from the ranked sample reproduces the expected structure of the main sequence. The luminous, O- and B- type stars occupy the upper left region, while the cooler F-, G-, K- and M- type stars extend towards redder colors indices and fainter magnitudes. The color coding spectral by class further confirms that the classifier partitions the parameter space in a manner consistent with established relationships, leading credibility to the ML approach.

5 Discussion

5.1 Physical Interpretation of Classifier Performance

The classification results are not just a statistical outcome- they represent the underlying physics of stellar atmospheres. The near-perfect categorization of O- and B- type stars is a direct consequence of their extreme effective temperatures, which drive their spectral energy distributions firmly into the ultraviolet and produce color indices ($g - r$) lead to

distinct cooler classes. In photometric spaces, these lead to occupy an ambiguous locus that even a simple classifier could recover.

The performance degraded across the F,G,K type stars, these classes spanned an effective temperature range of 3700-7500 K, over which the SDSS broadband filters sample a region of the Planck function that varies gradually. The $(g - r)$ index changed by less than 0.5 across the F-K sequence, compared to 1.5 across the entire range. The classifier is therefore operating near the intrinsic resolution limit of broadband photometry for these classes, and an F1 score of 0.67 for class F should be interpreted as a reflection of the physical information content available from the feature set.

5.2 Feature importance and physical interpretation

The dominance of the u and g bands in the importance analysis is physically consistent with theoretical stellar atmosphere models, as the u-band straddles the Balmer break and is highly sensitive to the ionized hydrogen. The classifier has evidently learned a meaningful temperature representation.

5.3 Limitations and Future Directions

The primary limitation of this approach is that broadband colors are degenerate with respect to surface gravity and metallicity- a G type dwarf and a G- type giant are nearly indistinguishable in the SDSS photometry despite differing so much more in their spectra.

Additionally, the oversampling of rare classes O and B by replication introduces no new information and might produce highly optimistic F1-scores for these classes. The 89.2% cross-validated accuracy nonetheless compares favorably with the inherent difficulty, given that a complete spectroscopic classification requires substantially greater investment per target. Future work incorporating DR3 parallaxes to derive absolute magnitudes, or extending the feature set with near infrared photometry from 2MASS [8] could improve discrimination in the F–G–K regime.

6 Conclusion

This study demonstrates that broadband SDSS, combined with an ensemble ML approach, is sufficient to classify stars along the Morgan-Keenan classification system with a cross validated accuracy of $89.2 \pm 1.2\%$. The Random Forest classifier successfully recovers the large scale temperature structure of the main sequence, obtaining near-perfect discrimination for the photometrically extreme O and B type stars while correctly identifying F–G–K regime as the intrinsic resolution limit of color data. The feature importance analysis confirms that the model has learned a meaningful representation of stellar effective temperature in which u and g bands dominated.

The result highlight both the strength and limitations of photometric classification at scale. While the approach cannot substitute for spectroscopic analysis in cases where surface gravity or metallicity are relevant, it offers an efficient and observationally inexpensive route to spectral class typing across large survey catalogs. As datasets from Gaia, LSST, and future wide field surveys continue to grow, ML methods trained on photometric features will become increasingly valuable tools for stellar studies. Incorporating parallax-derived absolute magnitudes and extending the wavelength baseline into the near infrared represent the most promising avenues for improving classification performance.

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